

FM214 Background Handout:

Present Value Mechanics and Statistics Review

FM214 Principles of Finance I • Week 0, self-study

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How to use this handout. This material is assumed known from Lecture 1 and is tested in Quiz 1. *It is not lectured.* Work through it before the course starts, do the practice exercises in Part 3, and come back to it whenever a lecture uses an annuity formula or a variance rule without comment.

Why a handout rather than two lectures? Earlier versions of this course spent their first two weeks on the material below: compounding, discounting, net present value, annuities and perpetuities, and a review of basic statistics. We no longer lecture it, for a simple reason: *the mechanics of discounting are easy, while the economics of risk and equilibrium are hard* — and the hard, interesting part deserves the lecture time. FM214 therefore opens directly with portfolio theory and the CAPM, and the present-value toolkit becomes background knowledge, exactly as basic calculus is background knowledge for your economics courses. Everything here is standard, largely timeless, and well within reach of self-study for a second-year student. None of it is optional, however: the valuation lectures (stocks, bonds, forwards, options) use these tools constantly and without re-derivation, Lecture 1 uses the statistics from Part 2 within its first hour, and Quiz 1 examines this handout directly.

What is in here. Part 1 develops present-value mechanics: future and present values, the net present value (NPV) rule and why it is the right rule, perpetuities and annuities with and without growth, compounding conventions and effective annual rates, and real versus nominal interest rates. Part 2 reviews the probability and statistics the course relies on: random variables, expectations, variances, covariances and correlations, the algebra for manipulating them (which is the engine of portfolio theory), and sample moments. Part 3 collects practice exercises with answers. Worked examples appear throughout in shaded boxes; treat every one of them as practice — cover the solution and attempt it yourself first.

Reading. Brealey, Myers, Allen and Edmans, *Principles of Corporate Finance* (14th edition, the course textbook): chapter 2 (“How to Calculate Present Values”) covers Part 1, and the section on real and nominal rates of interest in chapter 3 (“Valuing Bonds”) covers Section 6. Any introductory statistics text covers Part 2.

Part 1 Present Value Mechanics

Future values, present values, NPV, perpetuities and annuities, compounding conventions, real and nominal rates

1 Discount rates, compound interest, and future values

1.1 The discount rate

The primary concept underneath everything in this part is the **discount rate**: the reward that investors demand for accepting delayed rather than immediate gratification. We will also call the discount rate the *interest rate*, the *required rate of return*, or the *opportunity cost of capital*; the four terms are used interchangeably.

Why does such a reward exist? If you lend someone money for a year, you demand interest because you cannot instantly spend the money you have lent on consumption goods. If you lend money to a less trustworthy person or company, you require a greater interest rate because you are less confident that you will get your money back. The name “opportunity cost of capital” arises because the discount rate is also the return foregone by investing in a capital project rather than investing in freely-available securities.

Four objects recur throughout this part, and it is worth fixing them now. The *discount rate* is the interest rate used to compute present values of future cash flows. The *present value* is the value today of a future cash flow. The *future value* is the amount to which an investment will grow after earning interest. The *discount factor* is the present value of £1 received in the future. Each is defined precisely below.

Throughout Part 1 we assume (for now) a fixed, known interest rate of r per period. Later in the course — and this is where the interesting economics lives — we ask where r comes from and how it depends on risk.

1.2 Simple versus compound interest

We must distinguish between two varieties of interest calculation. Assume a one-period interest rate of r and an initial balance of X .

- **Simple interest:** the interest earned or paid is just the original balance of the deposit or loan times the interest rate. Over T periods, the total balance of your deposit or debt grows to

$$X + r \times X \times T = X(1 + rT).$$

- **Compound interest:** interest must be paid on previously charged or earned interest. Over T periods, the total balance of your deposit or debt grows to

$$X(1 + r)^T.$$

The difference is small over short horizons and enormous over long ones. Figure 1 shows the balance of £100 invested at 10% per annum under the two conventions: after 20 years simple interest delivers £300 while compound interest delivers roughly £673. Clearly, compound interest is good for savers but worrying for borrowers. Everything that follows uses compound interest.

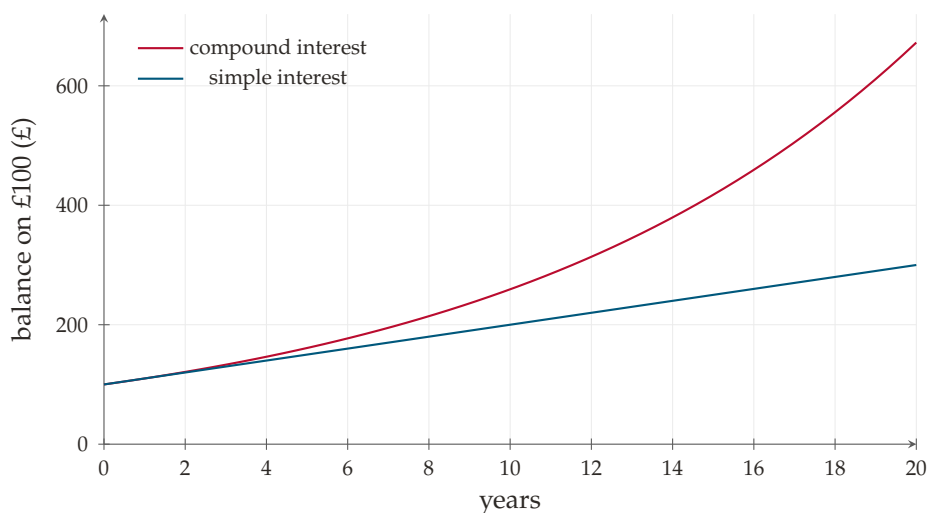


Figure 1: Compound versus simple interest at 10% per annum, initial balance £100.

1.3 Future values

Question: we are given a cash flow of X today. What value will this cash flow grow to if invested at interest rate r for T periods?

$$FV(r, T) = X(1 + r)^T$$

This amount is called the **future value** of X , assuming an interest rate of r and an investment period of T .

Three remarks. First, note the use of compound interest. Second, the future value is obviously larger if r is larger. Third, the future value is also greater when T is greater or when X is greater.

Example — future value of an inheritance

Scenario: you inherit £100,000 today and decide to place it in a bank account for 10 years. At the end of that 10-year period you will withdraw the funds and use the money to help buy your daughter a house.

- How much will you be able to contribute to her house purchase assuming that the annual interest rate is 5% for the next 10 years?
- Sensitivity analysis: how would your answer change if interest rates turn out to be 2% per year or 10% per year?

Solution: the baseline answer, assuming a 5% rate, and then the sensitivity analysis are given below;

$$FV(5\%, 10) = 100,000 \times (1 + 0.05)^{10} = 162,889.46$$

$$FV(2\%, 10) = 100,000 \times (1 + 0.02)^{10} = 121,899.44$$

$$FV(10\%, 10) = 100,000 \times (1 + 0.10)^{10} = 259,374.25$$

Note how sensitive a 10-year future value is to the rate: doubling the rate from 5% to 10% adds almost £100,000 to the outcome.

2 Present values and discount factors

2.1 Present values

Now run the logic in reverse. Assume that you are due to receive a payment of X in T years. What current cash flow is equivalent to that future cash flow? Compute the amount that you would have to place in a bank today such that it would grow to be exactly worth X in T years. Equivalently, ask how much would a bank lend you today if you promise to give them a repayment of X in T years. This is called the **present value** of X :

$$PV(r, T) = \frac{X}{(1 + r)^T}$$

Note the consistency check: if you took exactly this present value and banked it for T years at rate r you would end up with

$$PV(r, T) \times (1 + r)^T = X,$$

i.e. present values and future values are two views of the same equivalence.

2.2 Discount factors

When computing present values we often make use of an object called a **discount factor**. A discount factor is just the present value of £1. Discount factors vary with the interest rate and with the investment horizon. We calculate a discount factor as

$$DF(r, T) = \frac{1}{(1 + r)^T}.$$

Higher interest rates lead to lower discount factors, and longer investment periods lead to lower discount factors. Using the definition of the discount factor we can rewrite the present value compactly as

$$PV(r, T) = DF(r, T) \times X.$$

This notation earns its keep later in the course: bond prices, for example, are nothing but portfolios of discount factors.

Example — present value of a future receipt

Scenario: you are a supplier of aircraft parts to Boeing. In 2 years you are due to receive \$5,000,000 as payment for components you have supplied.

- How much is this worth in current terms assuming the current annual interest rate is 1%?
- Sensitivity analysis: how would your answer change if interest rates turn out to be 5% per year?

Solution: the baseline answer and then the sensitivity analysis are given below;

$$PV(1\%, 2) = 5,000,000 \times (1 + 0.01)^{-2} = \$4,901,480.25$$

$$PV(5\%, 2) = 5,000,000 \times (1 + 0.05)^{-2} = \$4,535,147.39$$

Clearly, even over short horizons, with reasonably high interest rates, present values can be much smaller than actual future cash flows.

3 Net present value and the NPV rule

3.1 From one cash flow to a stream

Until now we have thought about single cash flows (current or future) only. Most investments or applications involve multiple cash flows, received and paid at different points in time. Positive cash flows are receipts and negative cash flows are payments. How do you compute present values for these more complicated streams of money?

The solution is simple: take each individual cash flow and compute its present value, then sum present values across all of the cash flows. The result is called the **net present value**, or NPV. Receipts contribute positively to NPV and payments contribute negatively.

Formally, assume a constant interest rate of r per period, and an investment project which will deliver a cash flow of C_0 today, C_1 at the end of 1 period, C_2 at the end of two periods, continuing until it finishes by delivering a cash flow of C_K after K periods. The NPV is

$$NPV = \sum_{i=0}^K \frac{C_i}{(1+r)^i} = C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \cdots + \frac{C_K}{(1+r)^K}$$

Again, note that some of the C_i might be negative (if they are payments not receipts).

3.2 The NPV rule

The NPV rule: consider a financial or physical investment project for which you have calculated the NPV.

- If the NPV is positive, you should invest in the project.
- If the NPV is negative, you should turn down the investment opportunity.

Two comments. First, the discount rate used in the NPV calculation should reflect the project's risk: more risky projects require a greater return, and so you should use a larger discount rate. (Where that risk-adjusted rate comes from is precisely the subject of Lectures 1–3.) Second, if you do not know the cash flows associated with the project precisely, use the expected value of each cash flow instead.

Example — NPV of a property development

Scenario: a building company is considering buying a plot of land costing £2,000,000 today. It plans to build 5 houses on the plot. Construction will take 2 years to complete, with each house's building costs comprising £75,000 in one year and £40,000 in two years. If the company expects to sell each house for £600,000 two years from now and the interest rate is 8% per annum, should the project be taken on?

Solution: compute the NPV of the promised cash flow stream. This is

$$NPV = -2,000,000 - \frac{5 \times 75,000}{1 + 0.08} + \frac{5 \times (600,000 - 40,000)}{(1 + 0.08)^2} = 53,326.47.$$

As the NPV is positive, the company should go ahead with the project.

3.3 The optimality of the NPV rule

Why should you trust the NPV rule? Why is it optimal for individuals to invest in projects with positive NPV and discard projects with negative NPV? It turns out that NPV is optimal (under some assumptions) in the sense that use of the rule leads to investors maximising their expected wealth. This is true *regardless of how patient or impatient an investor is*, and thus the rule can be used for all investors: they will all agree on which investments to choose and which to discard. The following extended example makes the point.

Consider two investors:

- George (G) wants to consume now (he is impatient).
- Anne (A) wants to wait (she is patient).
- Both have current income of \$185,000 and expect zero income in a year.
- Each can invest in an opportunity costing \$185,000 now and returning a guaranteed \$210,000 at the end of the year.
- They also have access to a risk-free bank where the borrowing and lending rate is 5% per annum.

The claim: *both* should choose to undertake the investment.

Analysis of Anne's situation. Anne wants to consume at the end of the year. If she took her original wealth, invested it in the risk-free asset and consumed the proceeds, she would have $185,000 \times 1.05 = \$194,250$ to consume. Instead, if she invests in the project she has \$210,000 to consume at the end of the year. She is better off investing in the project.

Analysis of George's situation. George wants to consume today. He could just consume his original wealth of \$185,000. Instead, he could do the following: invest in the project, meaning that he has \$210,000 in income at the end of the year; borrow the present value of \$210,000 from the bank today — at 5% interest, this gives him \$200,000 today; consume the \$200k today, and in one year use the proceeds of the investment to repay the bank. He is better off investing in the project and using the capital market (i.e. borrowing and lending) to arrive at his desired consumption pattern.

George and Anne together. The outcome: both G and A end up investing in the project, as they are both better off if they do so. Why? It is because the project has positive NPV:

$$NPV = -185,000 + \frac{210,000}{1.05} = 15,000 > 0.$$

They should exploit the positive-NPV opportunity and then borrow or lend to move income through time and arrive at an optimal consumption pattern. Graphically: the project moves both investors onto a higher budget line in (dollars now, dollars next year) space — with slope $-(1 + r)$ — and each then picks their preferred point on it; Anne picks the point (\$0 now, \$210,000 next year), George the point (\$200,000 now, \$0 next year).

Implications (Fisher separation).

- Everyone, regardless of their preferences, should use the NPV rule to decide on which projects to invest in.
- The decision as to which projects to invest in can be *separated* from the decision regarding when to consume.

A self-assessment version of this scenario — with a *negative*-NPV project — is Exercise 1 in Part 3.

4 Perpetuities and annuities

Often the cash flow streams that we are interested in comprise many payments or receipts: stocks pay dividends on a regular basis; bonds make regular interest payments; after making a loan to a friend you expect monthly repayments. We can use PV techniques to arrive at a total value for these cash flow streams — but we can also use some shortcuts in certain cases to make our lives easier. The shortcuts are annuities and perpetuities, each with and without growth. These four formulas are used constantly in the rest of the course; the stock-valuation and bond-valuation lectures are essentially applications of them.

4.1 Perpetuities

Consider an asset that promises a fixed nominal cash flow at the end of every period from now until the end of time. This is a **perpetuity**. Call the cash flow C and assume that the one-period interest rate is fixed at r . Then the PV of the perpetuity is

$$PV_P = \sum_{i=1}^{\infty} \frac{C}{(1+r)^i} = C \sum_{i=1}^{\infty} \frac{1}{(1+r)^i}.$$

This infinite sum has a remarkably clean closed form:

$$PV_P = \frac{C}{r}$$

Example: if interest rates are at 5% per annum and someone promises you \$100 per year forever, that cash flow stream is worth $100/0.05 = \$2,000$ in present value terms.

Proof. Define x to be $(1+r)^{-1}$. Then the PV of the perpetuity is

$$PV_P = C \sum_{i=1}^{\infty} x^i = Cx \left(1 + \sum_{i=1}^{\infty} x^i \right).$$

Thus

$$x PV_P = Cx \sum_{i=1}^{\infty} x^i.$$

Combining these two equations gives

$$PV_P - x PV_P = Cx \left(1 + \sum_{i=1}^{\infty} x^i \right) - Cx \sum_{i=1}^{\infty} x^i = Cx,$$

and therefore

$$PV_P = C \frac{x}{1-x} = \frac{C}{r}. \quad \blacksquare$$

Perpetuities are not a textbook fiction: undated UK government debt. In late 2014 and early 2015 the UK government repaid undated (perpetual) bonds that had been issued by previous UK governments to finance, amongst other things, the First World War. The WW1 debt was, in total, around £1.9 billion with a 3.5% annual interest rate. The debt was perpetual and was originally issued in 1932. The undated debt was replaced with standard fixed-lifetime gilts issued at relatively low interest rates (given the economic conditions of the time). Plans were made and executed to take all undated UK bonds and replace them in the same way; some of these other undated bonds were first issued in the 18th century.

4.2 Annuities

More common are cash flow streams consisting of the payment of a fixed nominal cash flow, once per period, but only for a known, *finite* number of periods. This is an **annuity**. For example: I promise to pay my mortgage company £10,000 per year at the end of each year for the next 25 years.

The valuation intuition is simple, and clever. Take the example above:

1. Consider a perpetuity paying £10,000 per year with the first payment 1 year from now.
2. Consider a perpetuity paying £10,000 per year where the first payment is 26 years from now.
3. Subtract (2) from (1): you are left with a 25-year annuity.

Formally, assume a T -year annuity paying C per year and an annual interest rate of r . The value of a perpetuity paying C per year with the first payment in 1 year is

$$PV_1 = \frac{C}{r}.$$

The value of a perpetuity paying C per year with the first payment in $T + 1$ years is (a perpetuity worth C/r as of year T , discounted back to today)

$$PV_2 = \frac{1}{(1+r)^T} \frac{C}{r}.$$

The T -year annuity has value equal to PV_1 minus PV_2 :

$$PV_A = PV_1 - PV_2 = \frac{C}{r} - \frac{1}{(1+r)^T} \frac{C}{r} = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

The bracketed term (divided by r) is often called the *annuity factor*. Three worked examples follow, in increasing order of interest.

Example 1 — valuing an annuity product

Scenario: a recent retiree wants to use some of her savings to buy an annuity product. This product will deliver an annual income of \$25,000 per year for the next 15 years. (Assume that all payments are made at year end.) If the annual interest rate is currently 3% and we expect this rate to stay constant, how much should she expect to pay to the annuity provider today?

Valuation: the annual payment (C) is \$25,000, the constant interest rate (r) is 3% and the term (T) is 15 years. So

$$PV_A = C \left[\frac{1}{r} - \frac{1}{r(1+r)^T} \right] = 25,000 \left[\frac{1}{0.03} - \frac{1}{0.03 \times 1.03^{15}} \right] = \$298,448.38.$$

Example 2 — the 1998 Powerball prize

Scenario: in May 1998 a retired couple paid \$1 to buy a Powerball lottery ticket and won a record \$194 million. This sum was to be paid in 25 equal annual instalments. If the first instalment was received immediately and the interest rate was 9%, how much was the prize worth?

Valuation: the annual payment (C) is \$194 million divided by 25, which is \$7.76 million; the constant interest rate (r) is 9%; and the term (T) is 25 years.

Start by assuming that payments were made at the end of the year (so we can use the standard annuity formula):

$$PV_A = C \left[\frac{1}{r} - \frac{1}{r(1+r)^T} \right] = 7.76 \left[\frac{1}{0.09} - \frac{1}{0.09 \times 1.09^{25}} \right] = 76.22m.$$

But payments are actually made at the *beginning* of each year, so every payment arrives one year earlier than assumed, and the PV of the cash flows is

$$PV_A \times (1+r) = 76.22 \times 1.09 = 83.08m.$$

The headline “\$194 million” prize is worth \$83 million in present-value terms — less than half the advertised figure.

Example 3 — a 25-year mortgage (kept as practice — work it before reading on)

Scenario: assume a world where interest rates are expected to be constant at 3% per annum for the foreseeable future. You wish to buy a flat, and a mortgage company agrees to lend you £950,000 today. The mortgage has a term of 25 years. You promise the mortgage company a fixed repayment at the end of each of the next 25 years. What is the fair repayment?

Valuation: the repayment you promise is an annuity stream, and for it to be fair it must be worth £950,000 in PV terms. So we start with

$$PV_A = C \left[\frac{1}{r} - \frac{1}{r(1+r)^T} \right]$$

and this yields

$$950,000 = C \left[\frac{1}{0.03} - \frac{1}{0.03 \times (1.03)^{25}} \right].$$

Solving this, we see that the fixed annual repayment C is equal to £54,556.48.

Follow-up question: just after you have made your tenth mortgage payment, what is the size of your outstanding debt to the mortgage company?

After the tenth payment, you owe the mortgage provider fifteen more payments of £54,556.48, with the first payment one year away. The total PV of these payments, and thus the size of your outstanding mortgage debt, is

$$PV_A = 54,556.48 \left[\frac{1}{0.03} - \frac{1}{0.03(1.03)^{15}} \right] = 651,291.72.$$

Thus in the first 10 years of payments, you pay off roughly £300,000 of your £950,000 debt — much less than 10/25 of it, because early payments are mostly interest.

4.3 Perpetuities with growth

Another infinitely-lived stream of cash flows, this time growing over time. At the end of one period one is promised a cash flow of C , and at the end of each subsequent period the cash flow promised grows at rate g . Thus the second cash flow is $C(1+g)$, the third is $C(1+g)^2$, and so on. Again, if we assume that the constant one-period interest rate is r , then the PV of the **perpetuity with growth** is

$$PV_P = \sum_{i=1}^{\infty} \frac{C(1+g)^{i-1}}{(1+r)^i} = \frac{C}{r-g}$$

Proof. Start with the infinite sum:

$$PV_P = \sum_{i=1}^{\infty} \frac{C(1+g)^{i-1}}{(1+r)^i} = \frac{C}{1+r} \left[1 + \sum_{i=2}^{\infty} \left(\frac{1+g}{1+r} \right)^{i-1} \right] = \frac{C}{1+r} \left[1 + \sum_{i=1}^{\infty} x^i \right],$$

where $x = (1+g)/(1+r)$. This implies that

$$x PV_P = \frac{C}{1+r} \left[x + \sum_{i=2}^{\infty} x^i \right] = \frac{C}{1+r} \sum_{i=1}^{\infty} x^i.$$

Subtracting the second equation from the first we get

$$PV_P - x PV_P = \frac{C}{1+r} \left[1 + \sum_{i=1}^{\infty} x^i \right] - \frac{C}{1+r} \sum_{i=1}^{\infty} x^i = \frac{C}{1+r},$$

and finally

$$PV_P = \frac{1}{1-x} \frac{C}{1+r} = \frac{1+r}{r-g} \frac{C}{1+r} = \frac{C}{r-g}. \quad \blacksquare$$

Notes: clearly we need r to be greater than g to make the answer to this infinite present value sensible. If g is above r then the cash flows grow more quickly than the discount factors shrink, and thus the sum is infinite. (You will meet this formula again as the Gordon growth model for stock prices in Lecture 4.)

4.4 Annuities with growth

Finally, consider a cash flow stream such that at the end of one period one is promised a cash flow of C and at the end of each subsequent period the cash flow promised grows at rate g — but the stream *terminates* after T payments. This is an **annuity with growth**.

Valuation uses exactly the same trick as when valuing a no-growth annuity:

1. Value a perpetuity with growth where the first payment is made one period from now.
2. Value a perpetuity with growth where the first payment is made $T + 1$ periods from now (its first cash flow being $C(1 + g)^T$).
3. The cash flows from the annuity with growth equal the cash flows from (1) minus the cash flows from (2); thus its PV is the PV of (1) minus the PV of (2).

Valuation formula: if the first cash flow is C , the cash flows grow at rate g , and the one-period discount rate is r , then the PV of the growing annuity is

$$PV_A = \frac{C}{r-g} - \left(\frac{1+g}{1+r} \right)^T \frac{C}{r-g} = \frac{C}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^T \right]$$

Sanity checks: set $g = 0$ and you recover the ordinary annuity; let $T \rightarrow \infty$ (with $g < r$) and you recover the growing perpetuity.

5 Compounding conventions and effective annual rates

5.1 Quoted interest rates

Whenever we talk about interest rates we tend to scale them up or down so that they cover a period of exactly one year. The way in which this is done varies across financial instruments and markets. In this course we adopt the convention below, which is also common in real markets.

Quoted interest rates. The K -month interest rate is quoted at rate r :

- this rate r is *always* an annualised rate;
- the actual rate you will pay or receive on a K -month loan or deposit is $\frac{K}{12} r$.

Example: if someone quotes you an interest rate of 6% on a one-month deposit, this means that you will receive a rate of 0.5% on your initial balance at the end of the month. Why? Because $\frac{1}{12} \times 6\% = 0.5\%$.

5.2 Effective annual rates

The **effective (annual) interest rate** (EAR) is the actual interest you would receive on a deposit (or pay on a loan) on an annual basis, expressed as a percentage.

Example: take the situation above, where the quoted rate on a 1-month deposit is 6%. You make a deposit of \$1 and at the end of every month re-invest the balance at the 1-month rate. At the end of 12 months you have

$$1 \times (1 + r)^{12} = 1 \times (1 + 0.005)^{12} = 1.0617,$$

so your effective annual interest rate is 6.17%. The effective annual rate is *above* the quoted rate when the compounding period is less than a year: late in the year you earn interest on interest paid earlier in the year.

The table below (kept as practice — verify a few rows yourself) shows how the EAR on a quoted rate of 6% varies with the compounding frequency. The final row corresponds to compounding every hour ($8760 = 365 \times 24$ periods per year).

Quoted rate	Periods per year	Period interest	End-year value	EAR
6%	1	6%	1.06000	6.000%
6%	2	3%	1.06090	6.090%
6%	4	1.5%	1.06136	6.136%
6%	12	0.5%	1.06168	6.168%
6%	52	0.1154%	1.06180	6.180%
6%	365	0.0164%	1.06183	6.183%
6%	8760	0.000685%	1.061836	6.184%

Note two features: the EAR rises with the compounding frequency, but it converges rather than exploding — the gain from compounding hourly rather than daily is invisible at four decimal places. The limit is continuous compounding, treated below.

5.3 Terminology

When interest rates are quoted in this course, they will always be annualised rates. Here is how to decode phrases that will come up later on.

- **“The quarterly compounded interest rate is 5%”:** this is an annualised rate (as always); the actual rate that would apply to a 3-month deposit or loan is 1.25%; thus the annual effective rate is $1.0125^4 - 1 = 5.09\%$.
- **“The stated interest rate with monthly compounding is 9%”:** this is an annualised rate (as always); the actual rate that would apply to a 1-month deposit or loan is 0.75%; thus the annual effective rate is $1.0075^{12} - 1 = 9.38\%$.
- **“The semi-annually compounded interest rate is 7%”:** this is an annualised rate (as always); the actual rate that would apply to a 6-month deposit or loan is 3.5%; thus the annual effective rate is $1.035^2 - 1 = 7.12\%$.

5.4 Annuities under non-annual compounding

The annuity machinery of Section 4 works for any period length — months, quarters — provided the cash flow C , the rate r , and the term T are all expressed *per period*. Two worked examples.

Example — annuities and quoted rates

Scenario: an investment opportunity offers you quarterly cash flows of £1000 for the next 5 years. The interest rate is quoted at 4% with quarterly compounding. What is the maximum price you would pay to participate in this investment?

Solution: this is a non-growing annuity. We know that the PV of such a cash flow stream is

$$PV_A = C \left[\frac{1}{r} - \frac{1}{r(1+r)^T} \right],$$

where C is the periodic cash flow, r is the one-period interest rate and T is the number of periods. In our setting C is £1000. The quarterly interest rate is equal to the quoted rate divided by 4, i.e. 1%. The number of periods is the number of quarters in 5 years, i.e. 20. So

$$PV_A = 1000 \left[\frac{1}{0.01} - \frac{1}{0.01(1.01)^{20}} \right] = 18,045.55.$$

Example — a car loan (kept as practice — work it before reading on)

Scenario: you need £25,000 to buy a car today. You take out a 5-year loan to raise the money. The loan requires you to make constant monthly payments, with the first payment one month from today. If the interest rate with monthly compounding is 3% and we expect this rate not to move in the near future, what is the monthly payment you must make?

Solution: this is a standard annuity. We know that the PV of such a cash flow stream is

$$PV_A = \frac{C}{r} \left[1 - \frac{1}{(1+r)^T} \right].$$

In our case the PV of the repayments must be equal to £25,000, and the 1-month interest rate is $0.03/12 = 0.25\%$. There are 60 months in the payment period, so

$$25,000 = \frac{C}{0.0025} \left[1 - \frac{1}{(1.0025)^{60}} \right].$$

Solving, we find that our monthly repayment is £449.22.

Example — when payment frequency and compounding frequency differ

Question: you owe a supplier for materials. The payment schedule requires you to deliver £1000 to the supplier at the end of each of the next 9 months. What is the PV of this cash flow stream if the quarterly compounded interest rate is 7.5%?

First: compute effective annual and monthly rates. The effective annual rate is

$$r_A = \left(1 + \frac{0.075}{4}\right)^4 - 1 = 0.0771 = 7.71\%.$$

The effective monthly rate is

$$r_M = (1 + r_A)^{\frac{1}{12}} - 1 = 1.0771^{\frac{1}{12}} - 1 = 0.0062 = 0.62\%.$$

Finally: compute the PV of the 9-month annuity:

$$PV_A = 1000 \left[\frac{1}{0.0062} - \frac{1}{0.0062(1.0062)^9} \right] = £8,727.23.$$

The general principle: convert the quoted rate into an effective rate *per payment period*, then apply the standard annuity formula.

5.5 Continuous compounding

Fix a deposit of size X and an (annual) quoted interest rate of r . If there are m periods per year then, at the end of the year, the balance of one's investment is

$$X \left(1 + \frac{r}{m}\right)^m.$$

As the number of periods per year tends to infinity we get

$$\lim_{m \rightarrow \infty} X \left(1 + \frac{r}{m}\right)^m = Xe^r.$$

We call this **continuous compounding**, and r would be described as the *continuously compounded interest rate*. An investment for T years at continuously compounded rate r would grow to Xe^{rT} by the end of the holding period.

Continuous discounting: if you are due a cash flow of X in T years and the continuously-compounded interest rate is r , then the present value is

$$PV = Xe^{-rT}.$$

This convention reappears in the options material at the end of the course (the Black-Scholes formula is written with continuous discounting), so it is worth being comfortable with it now. It also explains the last row of the EAR table: with $r = 6\%$, $e^{0.06} = 1.0618$, matching the hourly-compounding entry to four decimal places.

6 Real and nominal interest rates

Consider the following three rates:

- **Inflation rate:** the (usually annual) rate at which the level of prices in the economy grows. Denote it by π .

- **Nominal interest rate:** the rate at which the balance of a deposit grows in *cash* terms. Denote it by r .
- **Real interest rate:** the rate at which the balance of a deposit grows in *purchasing power* terms. Denote it by i .

Up to now we have worked entirely in nominal terms. The discipline to remember: if you are given nominal cash flows, discount them at the nominal rate; if you are given real cash flows, you should discount them at the real rate. Mixing the two is one of the classic errors in valuation.

The Fisher relation. We define the real interest rate by

$$1 + i = \frac{1 + r}{1 + \pi}.$$

Approximate real interest rate. Rearranging,

$$(1 + r) = (1 + i)(1 + \pi) \Rightarrow 1 + r = 1 + i + \pi + i\pi.$$

When rates are low, the cross term $i\pi$ is very small. Ignoring it gives the familiar approximation

$$1 + r \approx 1 + i + \pi \Rightarrow \boxed{i \approx r - \pi}.$$

Comments. If the nominal interest rate is below the inflation rate, then a deposit is losing money in purchasing-power terms; on the flip side, a debt will shrink in real terms if $r < \pi$. Inflation rates can be substantial, implying that the difference between real and nominal rates can be large. For instance: assume nominal rates are 12% and inflation is running at 8%. The approximate real rate is $12\% - 8\% = 4\%$; the precise real rate is $1.12/1.08 - 1 = 3.7\%$. Note that the larger are the rates, the bigger the approximation error in the approximate calculation.

Part 2 Statistics Review

Random variables, moments, the algebra of means and variances, sample statistics

Risks and gambles are central to theoretical and practical finance: a share of stock *is* a gamble over next period's price and dividend. We will therefore rely on tools that describe gambles. This part collects every statistical fact the course uses. Lecture 1 applies this material immediately — the risk and return of a portfolio are exactly a mean and a variance — so make sure you can reproduce the boxed rules in Section 9 without looking.

An aside on notation. To avoid having to write out long sums or products of numbers, we will often make use of the following shorthand. **Summation:** if you want to write the sum of Z_i where i runs from 1 to K , we have

$$\sum_{i=1}^K Z_i = Z_1 + Z_2 + \cdots + Z_K.$$

Product: similarly, if you want to write the product of Z_i where i runs from 1 to K , we define

$$\prod_{i=1}^K Z_i = Z_1 \times Z_2 \times \cdots \times Z_K.$$

7 Random variables and probabilities

7.1 Discrete random variables

Setup: consider a random, *discrete* variable (RV) X_i :

- there are m possible outcomes, so $i = 1, 2, \dots, m$;
- define P_i to be the probability of X_i occurring;
- we must have $P_i \geq 0$ for all i ;
- we must also have $\sum_{i=1}^m P_i = P_1 + P_2 + \cdots + P_m = 1$.

A useful mental model for this course: the outcomes X_i are the possible returns on an asset next year, and the P_i are their probabilities.

7.2 Continuous random variables

When the set of possible outcomes for our RV is infinitely large, we must modify our thinking and our notation. We define $f(X)$ to be the **probability density function** (PDF) of the *continuous* RV X . It has the following properties:

- $f(X) \geq 0$ for all X ;
- $\int_{-\infty}^{\infty} f(X) dX = 1$;
- $\Pr(c \leq X \leq d) = \int_c^d f(X) dX$.

We define $F(X)$ to be the **cumulative distribution function** (CDF), where

$$F(X) = \int_{-\infty}^X f(Y) dY,$$

i.e. $F(x)$ is the probability that the random variable takes a value of at most x .

8 Population moments: mean, variance, covariance, correlation

8.1 Mean and variance

Take a continuous random variable X . Its population mean and variance are defined as

$$E(X) = \int_{-\infty}^{+\infty} X f(X) dX, \quad (1)$$

$$\text{Var}(X) = \int_{-\infty}^{+\infty} [X - E(X)]^2 f(X) dX, \quad (2)$$

where $f(X)$ is the probability density of X . (For a discrete RV, replace the integrals by sums: $E(X) = \sum_i P_i X_i$ and $\text{Var}(X) = \sum_i P_i [X_i - E(X)]^2$.)

The mean measures the “location” of the distribution of X , and the variance measures its “dispersion”. The square root of the variance is the **standard deviation**, written σ_X ; in finance, the standard deviation of a return is called its **volatility**. In this course: expected return will be our measure of the reward from an asset, and variance (or standard deviation) our first measure of its risk.

8.2 Covariance and correlation

Now assume that you have two variables X and Y . We define the covariance and correlation between them as

$$\text{Cov}(X, Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} [X - E(X)][Y - E(Y)] f(X, Y) dY dX, \quad (3)$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}, \quad (4)$$

where $f(X, Y)$ is the joint density of X and Y .

Both of these objects measure whether or not X and Y tend to move in the same direction or in opposite directions, relative to their respective means. The correlation is simply the covariance rescaled to be unit-free: it always lies between -1 and $+1$, with $+1$ meaning perfect positive linear dependence, -1 perfect negative linear dependence, and 0 no linear dependence. Covariance — not variance — turns out to be the central risk concept of this course: the CAPM’s message in Lecture 1 is that an asset’s price of risk depends on its covariance with the market, not on its own variance.

9 Rules for manipulating means, variances and covariances

The results in this section are used *again and again*: they are the entire mathematical content of portfolio theory. Assume that a, b and c are constants and X, Y and Z are random variables.

9.1 Means

The expectation operator is linear:

$$E(a) = a, \quad E(bX) = b E(X), \quad E(a + bX + cY) = a + b E(X) + c E(Y).$$

9.2 Variances and covariances

Manipulating variances and covariances: for constants a, b, c and random variables X, Y, Z ,

$$\begin{aligned}\text{Var}(a + X) &= \text{Var}(X) \\ \text{Var}(bX) &= b^2 \text{Var}(X) \\ \text{Var}(a + bX) &= b^2 \text{Var}(X) \\ \text{Cov}(X, X) &= \text{Var}(X) \\ \text{Cov}(a + X, b + Y) &= \text{Cov}(X, Y) \\ \text{Cov}(aX, bY) &= ab \text{Cov}(X, Y) \\ \text{Cov}(a, X) &= 0 \\ \text{Cov}(aX + bY + c, Z) &= \text{Cov}(aX, Z) + \text{Cov}(bY, Z) + \text{Cov}(c, Z) \\ &= a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)\end{aligned}$$

The intuitions are worth internalising. Adding a constant shifts a distribution without changing its dispersion, so it leaves variances and covariances alone. Scaling a variable by b scales deviations from its mean by b , so it scales the variance by b^2 (variance is in squared units) and covariances by b . A constant does not co-move with anything, so its covariance with any random variable is zero.

Finally, and probably most importantly:

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

This follows from the rules above, since $\text{Var}(aX + bY) = \text{Cov}(aX + bY, aX + bY)$; expand the covariance term by term.

9.3 The portfolio application

Here is where these rules will earn their keep from the first lecture onward. Suppose you split your wealth between two assets with random returns r_1 and r_2 , placing portfolio weights w_1 and w_2 on them (with $w_1 + w_2 = 1$). Your portfolio return is the random variable $r_p = w_1 r_1 + w_2 r_2$, and the rules above deliver immediately

$$\begin{aligned}E(r_p) &= w_1 E(r_1) + w_2 E(r_2), \\ \text{Var}(r_p) &= w_1^2 \text{Var}(r_1) + w_2^2 \text{Var}(r_2) + 2w_1 w_2 \text{Cov}(r_1, r_2).\end{aligned}$$

The expected return of a portfolio is the weighted average of expected returns, but its variance is *not* the weighted average of variances — the covariance term is where diversification lives. Lecture 1 starts from precisely these two equations.

10 Sample statistics

10.1 Sample mean and variance

The moments above are *population* quantities, defined from the true distribution. In practice we usually do not know the distribution from which X is drawn; instead we have data. Assume

that we have a set of N data points on a variable X . We will call the j th data point X_j , where j runs from 1 to N . The sample mean and variance are given by

$$\bar{X} = \frac{1}{N} \sum_{j=1}^N X_j,$$
$$\hat{\sigma}_X^2 = \frac{1}{N-1} \sum_{j=1}^N [X_j - \bar{X}]^2.$$

(The divisor $N - 1$ rather than N makes the sample variance an unbiased estimator of the population variance; for the sample sizes used in this course the distinction is numerically trivial.)

10.2 Sample covariance and correlation

If you have samples of size N for two variables, X and Y , then the sample covariance between them is given by

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{N-1} \sum_{j=1}^N [X_j - \bar{X}] [Y_j - \bar{Y}],$$

and the sample correlation is just the sample covariance divided by the square root of the product of the sample variances.

Useful Excel functions to compute sample statistics: make sure that you know how to use AVERAGE, STDEV, VAR, COVARIANCE.S and CORREL. When we compute historical average returns, volatilities and correlations from data in the course (and in the class exercises), these sample moments are what we are computing. (Use COVARIANCE.S, not COVAR: the latter divides by N rather than $N - 1$.)

Part 3 Practice Exercises

Attempt each exercise before reading its answer. Final answers with brief workings are given.

Exercise 1 is the self-assessment exercise from the original lecture deck; Exercises 2–5 are new to this handout. The worked examples in Parts 1–2 marked “kept as practice” (the mortgage, the car loan, and the EAR table) should also be treated as exercises: cover the solutions and reproduce them.

Exercise 1 (from the lecture deck’s self-assessment) The NPV rule with a bad project

Work through the George-and-Anne scenario of Section 3.3 under the assumption that the investment has a *negative* NPV: assume that the project pays off \$190,000, not \$210,000 (everything else unchanged: both investors have \$185,000 today, the project costs \$185,000, and the borrowing and lending rate is 5%). Show that both George and Anne should now turn the project down.

Answer. The NPV is now

$$NPV = -185,000 + \frac{190,000}{1.05} = -185,000 + 180,952.38 = -\$4,047.62 < 0.$$

Anne (patient): investing her \$185,000 at the bank gives $185,000 \times 1.05 = \$194,250$ next year, which beats the project’s \$190,000. She declines. *George* (impatient): if he took the project he could borrow at most its present value, $190,000/1.05 = \$180,952.38$, to consume today — less than the \$185,000 he can consume by simply not investing. He declines too. Both reject the project, whatever their preferences: the NPV rule again gives a unanimous verdict, this time negative.

Exercise 2 (new) Future values and compounding frequency

You deposit £20,000 for 8 years.

- If the annually compounded interest rate is 4%, what is the future value?
- If instead the *quarterly* compounded interest rate is 4%, what is the future value, and what is the effective annual rate?

Answer. (a) $FV = 20,000 \times 1.04^8 = £27,371.38$. (b) The quarterly rate is $4\%/4 = 1\%$ per quarter, and 8 years contain 32 quarters, so $FV = 20,000 \times 1.01^{32} = £27,498.81$. The EAR is $1.01^4 - 1 = 4.06\%$; more frequent compounding at the same quoted rate always leaves you (slightly) better off as a depositor.

Exercise 3 (new) A growing perpetuity

A share is expected to pay a dividend of £5 one year from now, and dividends are expected to grow at 2% per year forever after that. The relevant discount rate is 6% per annum.

- What is the present value of the dividend stream?
- What happens to your answer if the growth rate rises to 4%? To 6%?

Answer. (a) By the growing-perpetuity formula, $PV = C/(r - g) = 5/(0.06 - 0.02) = £125$. (b) With $g = 4\%$: $PV = 5/0.02 = £250$ — halving the gap between r and g doubles the value, so

values are very sensitive to growth assumptions. With $g = 6\% = r$ the formula breaks down: the infinite sum diverges, and the present value is unbounded. The formula requires $r > g$.

Exercise 4 (new) Comparing quoted rates; real versus nominal

- (a) Bank A offers a deposit rate of 8% with monthly compounding. Bank B offers 8.15% with annual compounding. Which deposit is better?
- (b) The better deposit earns its nominal return over a year in which inflation is 5%. What is the real return, exactly and approximately?

Answer. (a) Compare effective annual rates. Bank A: the monthly rate is $8\%/12 = 0.667\%$, so $EAR_A = (1 + 0.08/12)^{12} - 1 = 8.30\%$. Bank B: $EAR_B = 8.15\%$. Bank A is better despite the lower quoted rate — always compare EARs, never quoted rates. (b) With $r = 8.30\%$ and $\pi = 5\%$: the exact real rate is $i = 1.0830/1.05 - 1 = 3.14\%$; the approximation gives $i \approx 8.30\% - 5\% = 3.30\%$. The approximation overstates the real return, and the error grows with the level of the rates.

Exercise 5 (new) Portfolio mean and variance

Asset 1 has expected return $E(r_1) = 8\%$ and standard deviation $\sigma_1 = 20\%$; asset 2 has $E(r_2) = 4\%$ and $\sigma_2 = 10\%$; the correlation between their returns is $\rho_{12} = 0.25$. You invest 60% of your wealth in asset 1 and 40% in asset 2, so your portfolio return is $r_p = 0.6r_1 + 0.4r_2$.

- (a) Compute $E(r_p)$.
- (b) Compute $\text{Var}(r_p)$ and the portfolio standard deviation σ_p .
- (c) Compare σ_p with the weighted average of the two standard deviations, $0.6\sigma_1 + 0.4\sigma_2$. What explains the difference?

Answer. (a) $E(r_p) = 0.6 \times 8\% + 0.4 \times 4\% = 6.4\%$. (b) First convert the correlation to a covariance: $\text{Cov}(r_1, r_2) = \rho_{12} \sigma_1 \sigma_2 = 0.25 \times 0.20 \times 0.10 = 0.005$. Then

$$\text{Var}(r_p) = 0.6^2 \times 0.20^2 + 0.4^2 \times 0.10^2 + 2 \times 0.6 \times 0.4 \times 0.005 = 0.0144 + 0.0016 + 0.0024 = 0.0184,$$

so $\sigma_p = \sqrt{0.0184} = 13.6\%$. (c) The weighted average of standard deviations is $0.6 \times 20\% + 0.4 \times 10\% = 16\%$, noticeably above $\sigma_p = 13.6\%$. The gap is the benefit of *diversification*: because the assets are imperfectly correlated ($\rho_{12} < 1$), their movements partially offset, and portfolio risk is less than the average of the parts. This calculation is the seed from which the whole of Lecture 1 grows.

Checklist before Lecture 1. You should now be able to: move any cash flow through time with $(1 + r)^T$; state and defend the NPV rule; price perpetuities and annuities, with and without growth, from memory; convert any quoted rate into an effective rate per period; distinguish real from nominal rates via the Fisher relation; and compute the mean and variance of $w_1r_1 + w_2r_2$ without hesitation. If any item feels shaky, rework the relevant boxed examples — and remember that Quiz 1 examines this handout.